

Service Lifecycle Operations for Input

CYBOL Operation \ Channel	file directory	serial port	terminal	named pipeline	client socket	server socket	display
maintain/startup	-	+ (for cyboi client list)	+ (for cyboi client list)	+ (for cyboi client list)	-	+ (device specifics)	+
activate/enable	-	-	-	-	-	+	+ (global event loop)
dispatch/open	-	+ (device specifics)	+ (device specifics)	+ (device specifics)	+ (device specifics)	+ (in enable handler)	+ (window id)
feel/sense	-	+	+	+	-	+ (in enable handler)	+ (NO separate thread)
communicate/receive	+	+	+	+	+	+	+
communicate/send	+	+	+	+	+	+	+
feel/suspend	-	+	+	+		+ (automatically)	+ (automatically)
dispatch/close	-	+	+	+		+ (automatically)	+ (automatically)
activate/disable	-	-	-	-		+ (automatically)	+ (automatically)
maintain/shutdown	-	+	+	+		+	+
Display:							
open window manually so that its client entry contains a buffer							
catch events of ALL windows using global function of window system							
get id of window that the event stems from							
assign event to correct buffer of the corresponding client window id in cyboi							
use irq pipe to inform main thread							
read sender (client window id) from irq pipe in order to assign correct handler							
use client window id in handler in order to use the correct buffer of the client entry							
The feel/sense operation may be called for each window separately, in order to assign a specific handler (corresponds to the controller in MVC pattern).							
However, a new thread MUST NOT be started for the feel/sense function, since the global activate/enable catches all events.							
This can be achieved by distinguishing channels in the feel/sense cyboi function, to NOT create a thread for display.							
Serial Port and Terminal:							
The actually have no service running in the background.							
However, maintain/startup has to be called anyway in order to allocate server resources which contain a client list that is needed to manage the terminals.							
The identification numbers of client terminals and gui windows should NOT be managed in the cybol application.							
Instead, they should be added to the cyboi-internal client list.							
When a gui event occurs, it contains the corresponding window id so that the cybol app may use it (assigned to a cybol property handed over before).							
The feel/sense takes a handler as cybol property.							
Actually, dispatch/open could call feel/sense automatically in cyboi.							
But there are terminals that are just for output and do not need a sensing thread at all.							
Therefore, the feel/sense may be called optionally, but separately, after dispatch/open.							
Server Socket:							
In a first implementation version, the activate/enable within cyboi called dispatch/open plus feel/sense automatically.							
But to be more uniform to the other communication channels, this was changed so that they have to be called manually in a cybol handler now.							
Moreover, this is more flexible, since the handler for activate/enable may be adapted within cybol.							
The client socket id that was accepted in activate/enable gets handed over via pipe to the cyboi main thread and there it gets assigned to the cybol property „sender“.							
Within the activate/enable handler, the dispatch/open and feel/sense operations may relate to this cybol property „sender“, which is the client socket id.							
Client Socket:							
with or without thread or both possibilities							
if asynchronous sense + receive of messages, then thread is necessary and maintain/startup for client list as well							